

ORIGINAL ARTICLE

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Enhancing CO₂ hydrate kinetics with cost-effective calcium-based montmorillonite for carbon capture

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Abstract

CO₂ emission-driven climate change demands efficient CCUS technology. Montmorillonite (mmt) is an abundant, cheap, and eco-friendly terrestrial porous material, with the potential to turn from a passive geological medium to an active functional material for carbon capture. However, hydrate-based CO₂ capture, though promising for high storage density and eco-friendliness, is hindered by slow kinetics and high energy use. To address this, our study systematically explores the enhancement of CO₂ hydrate kinetics using mmt through both external optimization and intrinsic modification, by considering the effect of mmt concentration, SDS compounding, stirring rate, comparison with illite, and acid-leached modification. Results indicate that (i) mmt suspensions promote both the CO₂ hydrate nucleation and growth, particularly with low concentration, while the enhancement of mass transfer is crucial for the high concentration-mmt system. (ii) Subsequently, the addition of SDS (≥ 1000 ppm) can form micelles to enrich aqueous CO₂ locally, and the application of stirring (600 rpm) disrupted hydrate films to enable continuous CO₂ supply, which has both proved effective to promote CO₂ capture. (iii) The non-expandable illite performed worse than mmt, highlighting the expandable layers and high specific surface area for enhancing hydrate kinetics. This is further verified by acid leaching modification of mmt (m-mmt) that transforms its structure into a highly porous amorphous network. Notably, 1 wt% m-mmt achieved a CO₂ capture capacity of 93.9 mmol/mol, representing an 83.9% improvement over pristine mmt. These findings demonstrate the promising promotion effect of mmt for CO₂ hydrate, offering new insights for advancing cost-effective hydrate-based CO₂ capture.

Keywords Carbon capture, CO₂ hydrate, Montmorillonite, Illite, Acid leaching modification

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limitations, the stochastic nature of nucleation, inhomogeneous temperature and pressure fields within the reaction system, and the inherent incompleteness of the hydration reaction. This gap underscores the necessity of kinetic enhancement. Furthermore, the slow dissociation kinetics of CO₂ hydrates contribute to long-term structural stability, enhancing their potential for sustainable carbon sequestration [10–12].

Clay minerals are hydrous aluminosilicates with a layered structure, commonly including montmorillonite, kaolinite, and illite [13]. They are also typical porous materials which, like other well-known porous systems such as metal–organic frameworks [14], expanded graphite [15], artificial porous ice [16], and carbon nanotubes [17], can influence the kinetics of hydrate formation. The differences of various clays are detailed in Supplementary Information. Currently, studies of natural gas hydrate reservoirs have revealed that clay minerals in submarine sediments, particularly the most abundant montmorillonite and illite, play a critical regulatory role in the nucleation and stability of natural hydrates through their unique microscopic interfacial properties [18–20]. This natural phenomenon has inspired the application of hydrate technology in geological CO₂ storage [21–24]. As a result, attention has increasingly turned to the influence of such clay minerals on the formation process of CO₂ hydrates [25–27]. For instance, Feng et al. demonstrated that clay particles provide additional nucleation sites, thereby enhancing the conversion of water to hydrate, with montmorillonite increasing the water-to-hydrate conversion rate by 5.81% [28]. Lee et al. revealed that sodium-based montmorillonite reduces the reaction energy barrier for hydrate nucleation, significantly shortening induction time, which approximately followed a log-normal distribution under corresponding temperature and pressure conditions during the first 40 min of reaction [29]. However, during the growth stage of CO₂ hydrates, Ren et al. reported that sodium-based montmorillonite suspensions at 1–20 wt% concentrations exhibited lower CO₂ hydrate formation capacity compared to pure water systems, likely due to increased viscosity and consequent mass transfer limitations [30]. Yodpetch et al. reported that water adsorption on hydrophilic surfaces and interlayers of montmorillonite reduces capillary water flow in clay sediments. Hydrate blockage in small pores limits gas–water contact, thereby inhibiting CO₂ hydrate formation [31]. Furthermore, we have summarized the effects of various clay systems on the kinetics of CO₂ hydrate formation as reported across multiple studies (Table 1). Overall, these investigations reveal the following patterns: (i) Clay minerals can provide nucleation sites and significantly promote the nucleation stage of CO₂ hydrates. (ii) The influence of different clays on the

growth stage of CO₂ hydrates varies, likely due to differences in their surface properties. (iii) Low-concentration clay suspensions generally exhibit advantages in terms of hydrate formation kinetics.

Overall, the above studies have mainly focused on common clay minerals in the seafloor, namely montmorillonite and illite. Among them, montmorillonite (predominantly sodium-based) has received extensive attention due to its high specific surface area, high swelling capacity, and other characteristics. It is a 2:1 type clay mineral, composed of two tetrahedral silica sheets and one octahedral alumina sheet, with exchangeable cations present in the interlayer region. The key difference between calcium-based montmorillonite and sodium-based montmorillonite lies in the distinct interlayer cations, which further lead to variations in structure and surface properties. Recently, our studies reveal that in suspension systems, calcium-based montmorillonite (Ca-montmorillonite, note that the montmorillonite listed in Table 1 is sodium-based) facilitates CO₂ hydrate nucleation and growth, with low concentrations (≤ 5 wt%) demonstrating superior cost-effectiveness [19]. From these suspension-based experiments, it is evident that during the nucleation stage of CO₂ hydrates, the surface electric fields and abundant nucleation sites of these clays significantly shorten induction times. During the growth stage, Ca-montmorillonite, benefiting from its layered structure and hydration of Ca²⁺ ions, exhibits high growth rates even at low concentrations. Building on these findings, this study proposes a paradigm shift: repurposing these highly efficient natural promoters—particularly Ca-montmorillonite, which is abundant, easily accessible, and low-cost on land—from passive geological media into actively engineered, economical functional materials for artificial carbon capture processes. As a promoter for hydrate-based CO₂ capture technology, Ca-montmorillonite offers a compelling advantage, since most alternative materials suffer from either high synthesis costs, poor cost-effectiveness, or potential toxicity, unlike this naturally abundant and environmentally benign mineral.

The formation of CO₂ hydrates faces challenges for large-scale application due to high energy consumption and slow reaction kinetics, which must be addressed to improve economic viability and engineering feasibility. Current research commonly employs physical and chemical methods to enhance the formation rate of CO₂ hydrates [36, 37]. Physical enhancement aims to expand the gas–liquid contact area and improve mass transfer to promote hydrate formation, typically achieved through physical agitation [38–40]. Chemical enhancement involves thermodynamic promoters and kinetic promoters. Typical thermodynamic promoters include tetrahydrofuran (THF) [41], while kinetic promoters comprise

Table 1 Experimental materials in this study

No.	Ref.	Clay type	Initial condition	Solution	Clay content (wt%)	Kinetics	
						Nucleation	Growth
1	[32, 33]	Montmorillonite	274.15 K/3.0 MPa	Deionized water	3.3	Promotion	(1) Stage 1* rate increase (2) Stage 2* is suppressed
2					6.6		
3					10.0		
12	[28]	Montmorillonite	275.15 K/3.2 MPa	Seawater	1.0	Promotion	Water conversion rate increased by 1.12 times
13		Illite					Water conversion rate increased by 1.16 times
14	[30]	Montmorillonite	275.20 K/3.65 MPa	Deionized water	1.0	Induction time: 32 min (pure water: 61 min)	(1) Stage 1 rate increased by 32.13% (2) Stage 2 rate delayed by 30.72%
15					3.0	Induction time: 19 min	(1) Stage 1 rate increased by 88.75% (2) Stage 2 rate delayed by 18.76%
16					5.0	Induction time: 16 min	(1) Stage 1 rate delayed by 144.35% (2) Stage 2 rate increased by 6.10%
17					10.0	Induction time: 15 min	(1) Stage 1 rate increased by 134.85% (2) Stage 2 rate delayed by 28.84%
18					20.0	Induction time: 20 min	(1) Stage 1 rate delayed by 177.54% (2) Stage 2 rate increased by 3.05%
19					1.0	Induction period shortened by 67.92%	Gas conversion rate increased by 1.75 times
20	[34]	Bentonite	274.65 K/3.5 MPa	Seawater	3.0	Induction period shortened by 84.91%	Gas conversion rate increased by 2.24 times
21					5.0	Induction period shortened by 86.79%	Gas conversion rate increased by 2.06 times
22					7.0	Induction period shortened by 88.68%	Gas conversion rate increased by 1.93 times
23					1.0	Formation time prolonged by 176%	Inhibition
24						Formation time prolonged by 226%	
		Kaolinite					

* Stage 1: rapid growth stage after nucleation; Stage 2: Slow growth stage

surfactants and porous media, with common surfactants such as sodium dodecyl sulfate (SDS) being widely used [42]. The direct application of natural montmorillonite in carbon capture processes may face kinetic limitations such as prolonged induction times and slow growth rates. To address the above challenges, this study aims to develop a systematic enhancement strategy for montmorillonite performance.

Therefore, the main differences between this work and previous studies are as follows: (i) Inspired by the application of clay in hydrate-based carbon sequestration, we propose using these low-cost materials for CO₂ capture,

combined with mass transfer enhancement measures; (ii) We focus on calcium-based montmorillonite, which has received less attention in the field of CO₂ hydrate kinetics. For the first time, acid leaching modification of calcium-based montmorillonite is realized, and the regulation mechanism of its porous structure reconstruction on hydrate nucleation and growth is revealed; (iii) A dual-path enhancement strategy of “mass transfer enhancement (stirring/SDS)+intrinsic material modification (acid leaching)” is constructed, realizing the transformation from external field forced enhancement to intrinsic material enhancement. The research strategy

progressed through two complementary phases: First, conventional enhancement methods were examined, including mass transfer intensification (e.g., increased stirring rate) and surfactant compounding (e.g., SDS), to achieve rapid kinetic improvement. Subsequently, the focus shifted to material-centric strategies, exploring the role of intrinsic clay properties—comparing expandable versus non-expandable structures—and implementing acid-leaching modification to fundamentally reconstruct montmorillonite's structure. The modification resulted in substantially enhanced specific surface area and hierarchical pore channels, far exceeding the natural state of the clay and elevating its promotion performance to unprecedented levels. These findings can provide new solutions for sustainable carbon capture technology based on natural minerals.

2 Methods

2.1 Materials and experimental apparatus

In this study, the materials, including CO₂ gas, deionized water, Ca-montmorillonite (mmt), illite, sodium dodecyl sulfate (SDS), and concentrated hydrochloric acid (HCl) (Table 2) were used to analyze the kinetics of CO₂ hydrate formation.

The experimental apparatus used in this study is shown in Fig. 2. The reactor is made of 316 stainless steel, with an effective volume of 259.9 mL. Three sapphire windows with a diameter of 30 mm are installed on the front and rear sides of the reactor and on the top. During the experiment, the formation process of CO₂ hydrate can

be recorded by a CCD camera. In addition, two temperature sensors (PT-100, ± 0.1 K) and a pressure sensor (± 0.01 MPa) are installed on the side of the reactor to measure the temperature of the gas phase and the liquid phase, respectively, which are connected to the computer through the data acquisition system to obtain real-time temperature and pressure information. The reactor is located in the water bath system, and the temperature inside the reactor is controlled by adjusting the external water temperature.

2.2 Experiment procedures

2.2.1 Acid leaching modification experiment

Typically, mmt possesses a layered structure, demonstrating potential for acid leaching modification [43]. This method can increase the specific surface area and pore volume of mmt, creating a structure more conducive to hydrate growth and thereby enhancing its storage capacity. The acid leaching modification scheme adopted in this study is illustrated in Fig. 3a, utilizing an optimal HCl concentration of 6 mol/L for the modification [44]. The specific experimental procedure for modification is as follows: (i) Weigh high-purity (99.9%) calcium-based montmorillonite and mix it with 6 mol/L hydrochloric acid at a mass ratio of 1:10; (ii) Place the mixture in an oil bath at 95 °C with continuous stirring for 4 h; (iii) Rinse with deionized water and perform suction filtration. Check the filtrate using pH test paper; continue filtration until the pH test indicates neutrality; (iv) Dry the sample in an oven at 90 °C for 10 h until completely dry. SEM images reveal that the original mmt exhibits a highly compact layered structure with a relatively smooth surface, limited porosity, and no obvious pore network (Fig. 3a1). In contrast, the modified mmt (m-mmt) shows significantly increased surface roughness, disruption of the original dense layered structure, and the formation of a highly dispersed porous network, resulting in a markedly more open and loose structure. Furthermore, the PXRD patterns in Fig. 3b confirm the successful modification, indicating destruction of the crystal plane structure in mmt after modification, leading to reduced crystallinity and an increase in defects.

Furthermore, the modified m-mmt exhibits improved properties in several aspects. Thermogravimetric (TG) analysis (Fig. 3c) shows residual rates of 79.41% for pristine mmt and 90.77% for m-mmt, indicating that the hydrochloric acid modification removed impurities within the montmorillonite samples, thereby enhancing both purity and thermal stability of m-mmt. Additionally, nitrogen adsorption–desorption isotherm tests (Fig. 3d) conducted on samples before and after modification reveal that m-mmt demonstrates higher adsorption capacity across the relative pressure (P/P_0) range compared to unmodified mmt. This

Table 2 Experimental materials in this study

Material	Parameter	Source
CO ₂ gas	Purity 99.9%	Shenzhen Huateng Gas Co., Ltd
Deionized water	18.2 MΩ·cm	Deionized water purification system
Mmt	Purity 99.9%	Hubei Sand Technology Co., Ltd
Illite	Purity 99.9%	Shanlin Poetry Minerals Company
SDS	Purity 99.5%	Shenzhen Kezhubao Biotechnology Co., Ltd
HCl	36.0–38.0%	Shenzhen Chemical Technology Co., Ltd

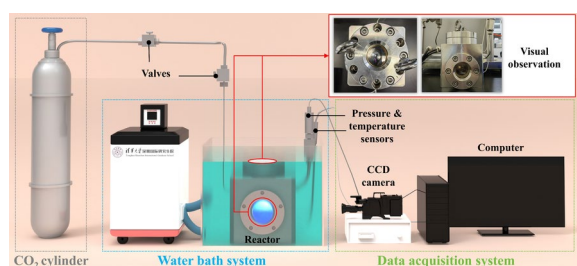


Fig. 2 Schematic of the experimental apparatus

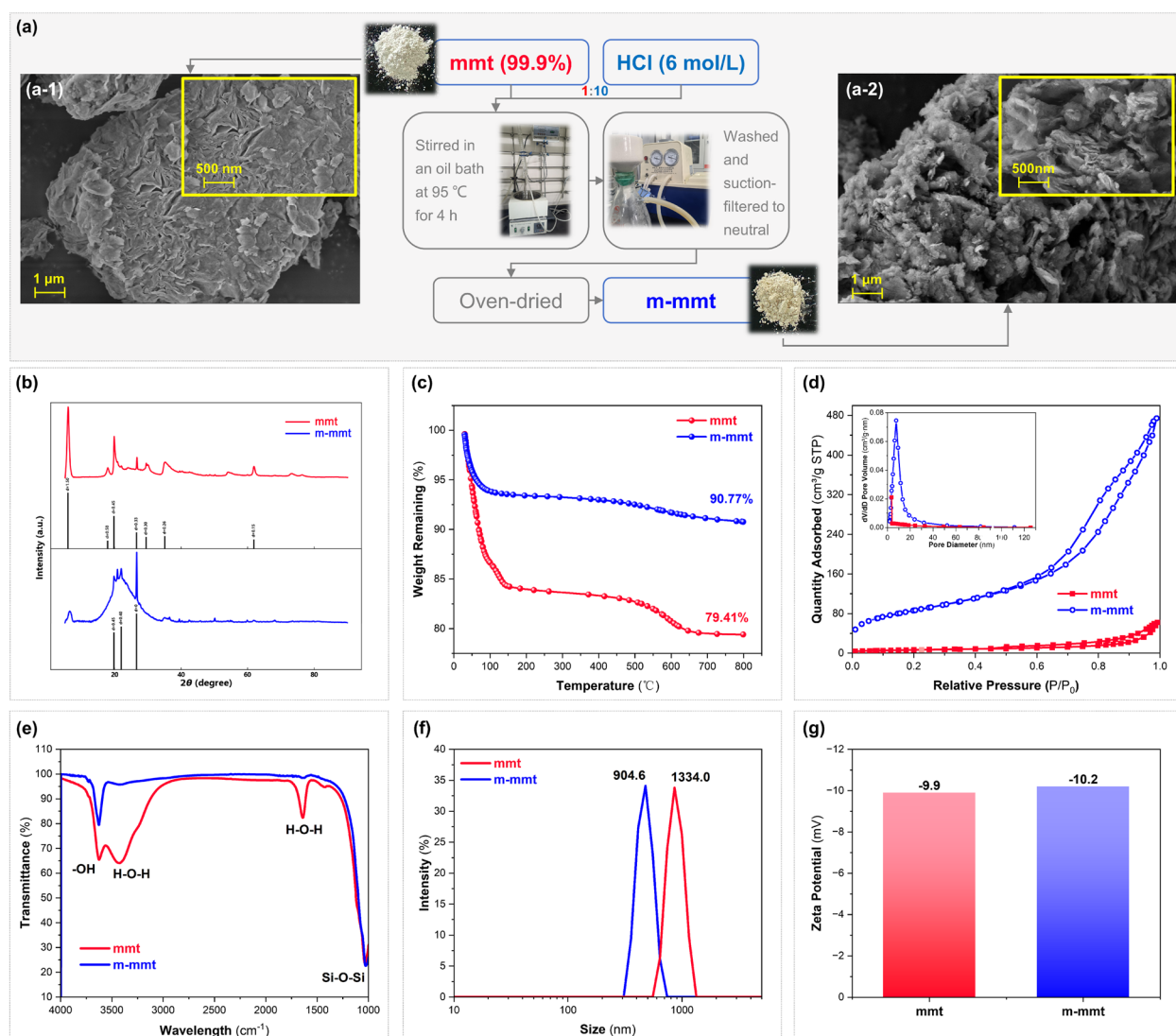


Fig. 3 Acid leaching modification: (a) Experimental process and SEM images; (b) The PXRD patterns; (c) Thermogravimetric results; (d) Nitrogen adsorption–desorption curves; (e) The FTIR patterns; (f) Size distributions of 1 wt% mmt and m-mmt systems; (g) Zeta potential of 1 wt% mmt and m-mmt systems

suggests that m-mmt possesses either a higher specific surface area or greater pore volume. The pore size distribution plots (inset) illustrate the characteristics of both materials. Calculated from the desorption branch using the cylindrical pore model with Kelvin equation and thickness curve corrections, the mmt shows a unimodal distribution, indicating a uniform mesoporous structure with an average pore size of 12.05 nm. In contrast, the m-mmt displays a broad peak with an average pore size of 9.23 nm, indicating that it forms a more complex hierarchical pore structure while retaining the dominant mesoporous structure. As summarized in Table 3, acid treatment resulted in expanded interlayer spacing and increased surface active sites in the m-mmt. The BET specific surface area of the m-mmt became 13 times

greater than that of the original material, demonstrating a significant increase in exposed surface area. Combined with FTIR and XRD results (Fig. 3b and e), the peaks of hydroxyl groups and Al–O bonds weakened during acid leaching. HCl dissolved interlayer metal cations of montmorillonite,

Table 3 Comparative structural parameters of pristine and acid-modified montmorillonite

Sample	Pore volume (cm³/g)	BET surface (m²/g)	Aperture (nm)
mmt	0.09	23.54	12.05
m-mmt	0.73	312.12	9.23
Change degree	+711.11%	+1226.08%	-23.40%

destroying its compact layered structure and crystal planes and forming abundant mesopores and a small number of micropores. This is also observed in SEM morphologies before and after acid leaching (Fig. 3a), showing layered structure disintegration and mesoporous channel formation. In addition, from the size distribution (Fig. 3f) and Zeta potential analysis (Fig. 3g), acid-leached m-mmt has a significantly broader particle size distribution with an overall reduction in particle size and has a lower isoelectric point than pristine mmt, indicating that acid leaching changes the material's surface morphology, modifies its particle size distribution, and enhances dispersion. These characterizations confirm the success of acid leaching modification.

2.2.2 Kinetics experiment

In this study, the experimental setup shown in Fig. 2 and the materials listed in Table 2 were used. The procedure was conducted as follows: (i) The reactor was cleaned with deionized water and dried. (ii) An 80 g suspension with a clay concentration of 0–5 wt% was loaded into the reactor, which was then sealed. (iii) The water bath temperature was set to 293.15 K, and residual air was removed from the reactor by vacuum pumping. (iv) CO₂ gas was injected into the reactor at a constant rate until the pressure reached 4.5 MPa. The valve was then closed, and magnetic stirring was initiated to enhance CO₂ dissolution. (v) After 3 h of stirring, the water bath temperature was gradually cooled to 275.15 K. The reaction system was maintained at this temperature for 15 h to monitor and document hydrate formation.

Through the described procedures, temperature, pressure, and visual images were directly recorded by the data acquisition system and the CCD camera. The evolution of temperature and pressure throughout the kinetic experiment is shown in Fig. 4 and can be divided into six main stages: (i) Vacuum pumping stage (t_0 – t_1); (ii) CO₂ injection stage (t_1 – t_2); (iii) CO₂ dissolution stage (t_2 – t_3); (iv) Cooling nucleation stage (t_3 – t_5); (v) CO₂ hydrate formation stage (t_5 – t_6); (vi) Heating decomposition stage (t_6 –). In this regard, we defined the following indicators to quantify its kinetic performance.

(1) Induction time

Induction time (t_{ind}) is defined as the duration between the temperature and pressure conditions of the reactor entering the hydrate phase equilibrium zone (Fig. 4, t_4) and the CO₂ hydrate formation (t_5) observed in the sapphire windows.

$$t_{ind} = |t_4 - t_5| \quad (1)$$

(2) CO₂ solubility

CO₂ solubility (x) is calculated by Eqs. (2) and (3):

$$x(T, P) = \frac{f(T, P)}{H^* \times \exp\left(\left(v_{CO_2}^\infty P\right)/RT\right)} \quad (2)$$

$$H^*(T) = \exp\left(-6.8346 + 1.2817 \times \frac{10^4}{T} - 3.7668 \times \frac{10^6}{T^2} + 2.997 \times \frac{10^8}{T^3}\right) \quad (3)$$

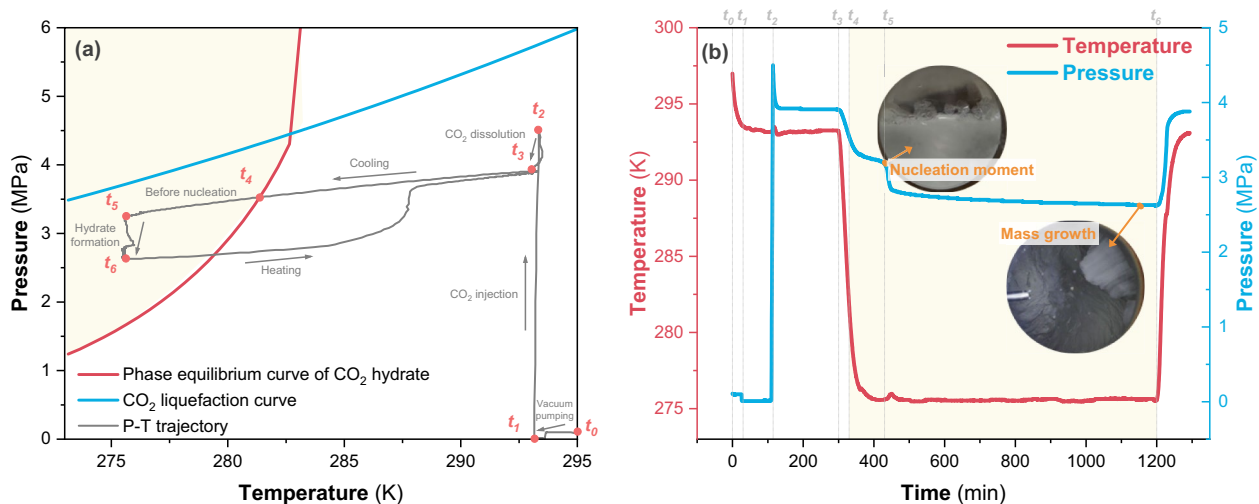


Fig. 4 Temperature and pressure (P–T) evolution for the kinetics experiment process: (a) P–T changes with time; (b) P–T trajectory

where f is the fugacity of CO_2 at experimental temperature (T) and experimental pressure (P) in the absence of hydrates, and its value is calculated by the SRK equation of state [45]; when hydrate exists, the pressure adopts gas–liquid–solid three-phase equilibrium pressure; H^* is the Henry's law constant of CO_2 [46]; v_{CO_2} represents the partial molar volume of CO_2 in an infinitely diluted aqueous solution, with a value of $3.32 \text{ cm}^3/\text{mol}$ [47]. R is the universal gas constant, with a value of 8.314 .

(3) CO_2 hydrate formation amount

The molar amounts of CO_2 in the gas phase, aqueous phase, and hydrate phase during hydrate formation are quantified using the volumetric method, with the specific expressions given by Eqs. (4) to (9):

$$V = V_{\text{CO}_2,t}^H + V_{\text{H}_2\text{O},t}^A + V_{\text{CO}_2,t}^G \quad (4)$$

$$V_{\text{CO}_2,t}^H = n_{\text{CO}_2,t}^H \cdot v_{\text{CO}_2}^H \quad (5)$$

$$V_{\text{H}_2\text{O},t}^A = V_{\text{H}_2\text{O},0}^A - N_H n_{\text{CO}_2,t}^H v_{\text{H}_2\text{O},t}^H \quad (6)$$

$$V_{\text{CO}_2,t}^G = \left(n_{\text{CO}_2,t}^G \right) \frac{Z_t R T_t}{P_t} = \left(n_{\text{CO}_2,0}^G - n_{\text{CO}_2,t}^H - n_{\text{CO}_2,t}^A \right) \frac{Z_t R T_t}{P_t} \quad (7)$$

$$n_{\text{CO}_2,t}^A = x_t \times n_{\text{H}_2\text{O},t}^A = x_t \times \left(n_{\text{H}_2\text{O},0}^A - N_H \times n_{\text{CO}_2,t}^H \right) \quad (8)$$

$$v_{\text{CO}_2}^H = \frac{M_{\text{CO}_2}}{\rho_{\text{CO}_2}^H} \quad (9)$$

where: The reactor volume (V) and the initial volume and molar amount of water are known prior to the experiment. In these symbols, superscripts H , A , and G represent the hydrate phase, aqueous phase, and gas phase, respectively. Subscripts 0 and t denote the initial time and time t . For instance, the temperature and pressure at the initial moment (T_0 and P_0), and at a specific time (T_t and P_t), are collected by the data acquisition system. Based on the real-time temperature and pressure variations in the reactor, the CO_2 solubility (x) and compressibility factor (Z , calculated by the Peng–Robinson equation of state) can be calculated by Eqs. (2) to (3). Other parameters are adopted according to previous studies and empirical values as follows: The hydration number is taken as $N_H = 6.2$, assumed [48]. The molar mass of CO_2 (M_{CO_2}) is 44 g/mol , and the density of pure CO_2 hydrate ($\rho_{\text{CO}_2}^H$) is taken as 1.1 g/cm^3 [49]. The molar volume of water ($v_{\text{H}_2\text{O}}$) is a constant value of $18.0 \text{ cm}^3/\text{mol}$, and $v_{\text{CO}_2}^H$ is the molar volume of CO_2 hydrate. The theoretical value of N_H is 5.75 , but it is usually set between 6 and 6.5 in experiments

due to practical limitations. The parameter x is ignored in some studies; when dissolution is considered, the calculated CO_2 hydrate yield becomes relatively conservative. Thus, by measuring the temperature and pressure changes in the reactor, key indicators, including the CO_2 dissolution amount and CO_2 hydrate formation amount, can be obtained.

(4) Capture increase rate

Capture increase rate (R_c) is defined as the increase in real-time CO_2 capture capacity of a case compared to a control case.

$$R_c = \frac{n_{\text{CO}_2,t}^H(\text{Case}) - n_{\text{CO}_2,t}^H(\text{Ref})}{n_{\text{CO}_2,t}^H(\text{Ref})} \times 100\% \quad (10)$$

2.2.3 Experimental cases

To investigate the kinetics of CO_2 hydrate formation and promotion measures using clay-based media, the following cases were designed (Table 4): (i) Cases 0–1 and 0–2 serve as baseline cases, providing a reference for other enhanced cases. (ii) Cases 1–1 to 1–4 use varying concentrations of mmt as the variable, aiming to explore the effect of the clay-based medium under low mass transfer conditions (400 rpm). (iii) Cases 2–1 to 2–4 combine a fixed concentration ($1.0 \text{ wt\%}$) of mmt with SDS, aiming to investigate the synergistic promotion effect on hydrate formation. (iv) Cases 3–1 to 3–4 examine the effect of altered mass transfer conditions (600 rpm), aiming to explore the differences compared to low mass transfer conditions (Cases 1–1 to 1–4). (v) Cases 4–1 to 4–4 involve a change in clay type (to illite), aiming to compare its performance with mmt (Cases 3–1 to 3–4) under high mass transfer conditions. (vi) Cases 5–1 to 5–4 utilize acid-leached m-mmt, aiming to further exploit the material's performance. In addition, we conducted recyclability tests for Case 5–2. The main procedure of this test is as follows: m-mmt was reused; after each hydrate formation cycle, the temperature of the entire reactor was raised to decompose the hydrates, followed by thorough stirring at a high temperature for a certain period of time to eliminate potential memory effects. Subsequently, the temperature was lowered to initiate hydrate formation again, and this process was repeated cyclically.

In this work, the concentration of the suspension was controlled within $0.5\text{--}5 \text{ wt\%}$, mainly based on the following considerations: (i) Previous studies have confirmed that clay minerals exhibit the optimal promoting effect on hydrate kinetics at low concentrations. (ii) A low-concentration clay system is more favorable for industrial slurry transportation and stirring operations, which

Table 4 Cases design for the experiment

No.	Medium	Stirring rate	SDS concentration	Purpose
Case 0–1	Pure water	400 rpm	/	Reference cases
Case 0–2	Pure water	600 rpm	/	
Case 1–1	0.5 wt% mmt	400 rpm	/	Clay-based medium effects
Case 1–2	1.0 wt% mmt	400 rpm	/	
Case 1–3	3.0 wt% mmt	400 rpm	/	
Case 1–4	5.0 wt% mmt	400 rpm	/	
Case 2–1	1.0 wt% mmt	400 rpm	500 ppm	Compound SDS effects
Case 2–2	1.0 wt% mmt	400 rpm	1000 ppm	
Case 2–3	1.0 wt% mmt	400 rpm	2000 ppm	
Case 2–4	1.0 wt% mmt	400 rpm	4000 ppm	
Case 3–1	0.5 wt% mmt	600 rpm	/	Stirring rate effects
Case 3–2	1.0 wt% mmt	600 rpm	/	
Case 3–3	3.0 wt% mmt	600 rpm	/	
Case 3–4	5.0 wt% mmt	600 rpm	/	
Case 4–1	0.5 wt% illite	600 rpm	/	Clay types effects
Case 4–2	1.0 wt% illite	600 rpm	/	
Case 4–3	3.0 wt% illite	600 rpm	/	
Case 4–4	5.0 wt% illite	600 rpm	/	
Case 5–1	0.5 wt% m-mmt (modified)	400 rpm	/	Acid modification effects
Case 5–2	1.0 wt% m-mmt (modified)	400 rpm	/	
Case 5–3	3.0 wt% m-mmt (modified)	400 rpm	/	
Case 5–4	5.0 wt% m-mmt (modified)	400 rpm	/	

aligns with the cost-effective and high-efficiency design philosophy of this study. All experimental cases in this study were conducted with 3 parallel replicate experiments to ensure the reliability and repeatability of kinetic data. The key kinetic results (induction time, final CO₂ capture capacity, average hydrate growth rate) and initial experimental conditions (temperature, pressure, stirring rate) of all cases have been systematically compiled in Table S1 of the Supplementary Information, with the average value and standard deviation of three replicates listed for intuitive display of experimental trends.

3 Results analysis

3.1 Clay-based medium effects

First, we investigated the kinetic effects of mmt suspensions at different concentrations on CO₂ hydrate formation under low mass transfer conditions (400 rpm). Figure 5a shows the differences in induction time, revealing that a certain amount of mmt can promote hydrate growth in the initial stage by providing nucleation sites. The induction time further shortens with increasing concentration, which is primarily related to the surface properties of mmt and enhanced mass transfer. The evolution of the capture rate from 180 to 300 min (Fig. 5b) indicates that high-concentration mmt suspensions exhibit the fastest reaction rates during the initial nucleation phase,

while this effect is less pronounced in low-concentration (0.5 wt%) and pure water systems. This observation is consistent with their induction times (Fig. 5a), where pure water and the 0.5 wt% system show longer induction times compared to other concentrations.

Furthermore, Fig. 5c illustrates the dynamic differences in normalized CO₂ capture capacity (CO₂ hydrate amount/water amount, mmol/mol). The black dashed lines in the figure mark the variation of dissolved CO₂ in the aqueous phase, which reaches a peak of approximately 30 mmol/mol within the initial 180 min. It can be observed that the initial 180-min period is primarily dominated by CO₂ dissolution. Subsequently, as the system enters the hydrate phase equilibrium zone, the dissolved CO₂ in the aqueous phase serves as a critical precursor for hydrate formation and is continuously consumed by combining with water molecules at the nucleation sites provided by montmorillonite, resulting in a gradual decrease in the dissolved CO₂ concentration. Dissolved and gaseous CO₂ combine with water to form hydrates, leading to a further increase in CO₂ capture. The final results show that the 0.5 wt% system achieves the maximum CO₂ capture capacity of 69.2 mmol/mol. This confirms that an appropriate amount of clay particles creates obstruction in the stirred solution, resulting in more intense agitation and enhanced mass transfer,

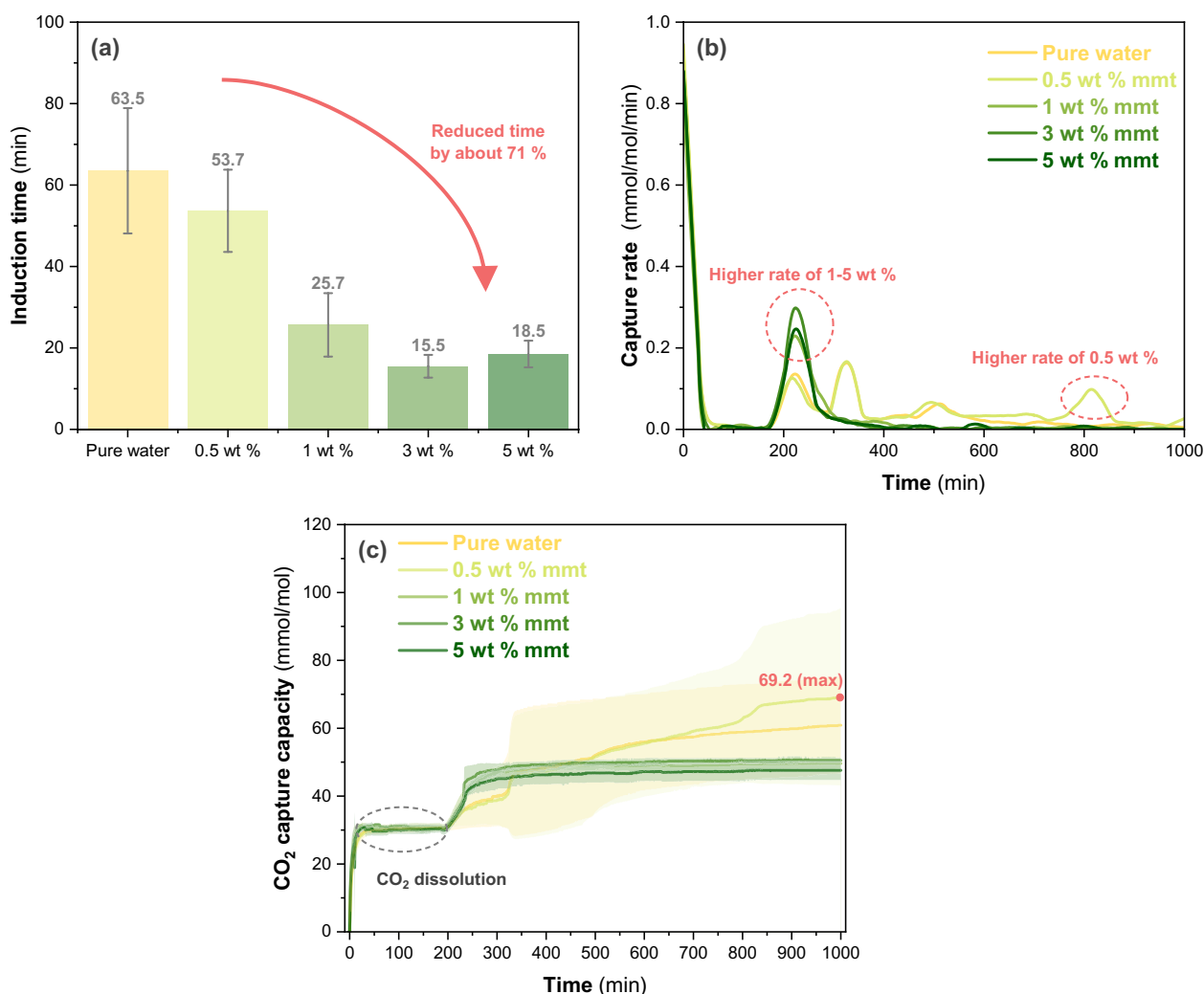


Fig. 5 CO₂ capture characteristics with low mass transfer in montmorillonite (mmt) system: (a) Induction time; (b) Capture rate; (c) CO₂ capture capacity

particularly in the later stages of the reaction, where low-concentration systems still maintain relatively high capture rates (Fig. 5b). In contrast, high-concentration systems demonstrate inferior capture capacity compared to low-concentration systems. This may be attributed to the high viscosity of high-concentration suspensions and the strongly polarized water layers induced by the clay, which hinder mass transfer of CO₂ from the gas phase to the liquid phase [30]. Under the same external mass transfer conditions, the rapid initial hydrate growth, combined with the inherently higher viscosity of high-concentration suspensions, significantly increases mass transfer resistance due to this dual effect. Consequently, magnetic stirring becomes less effective, and the hydrate film formed at the gas–liquid interface cannot be disrupted, limiting further CO₂ capture. This aspect will be

discussed in detail in the following section on mass transfer intensification.

3.2 Reinforcing the mass transfer intensification process

The typical surfactant SDS is widely used as a kinetic promoter for hydrate formation. Its composite system with mmt is expected to further enhance CO₂ capture capacity through synergistic effects. To investigate this, we prepared composite systems using 1 wt% mmt with SDS solutions at concentrations ranging from 500 to 4000 ppm and conducted comparative experiments under low mass transfer conditions (400 rpm). As shown in Fig. 6a, a significant reduction in induction time is observed once the SDS concentration reaches or exceeds 1000 ppm. This can be attributed to the fact that SDS generally forms abundant micelles in solution, which effectively concentrate dissolved CO₂ molecules, substantially increasing the

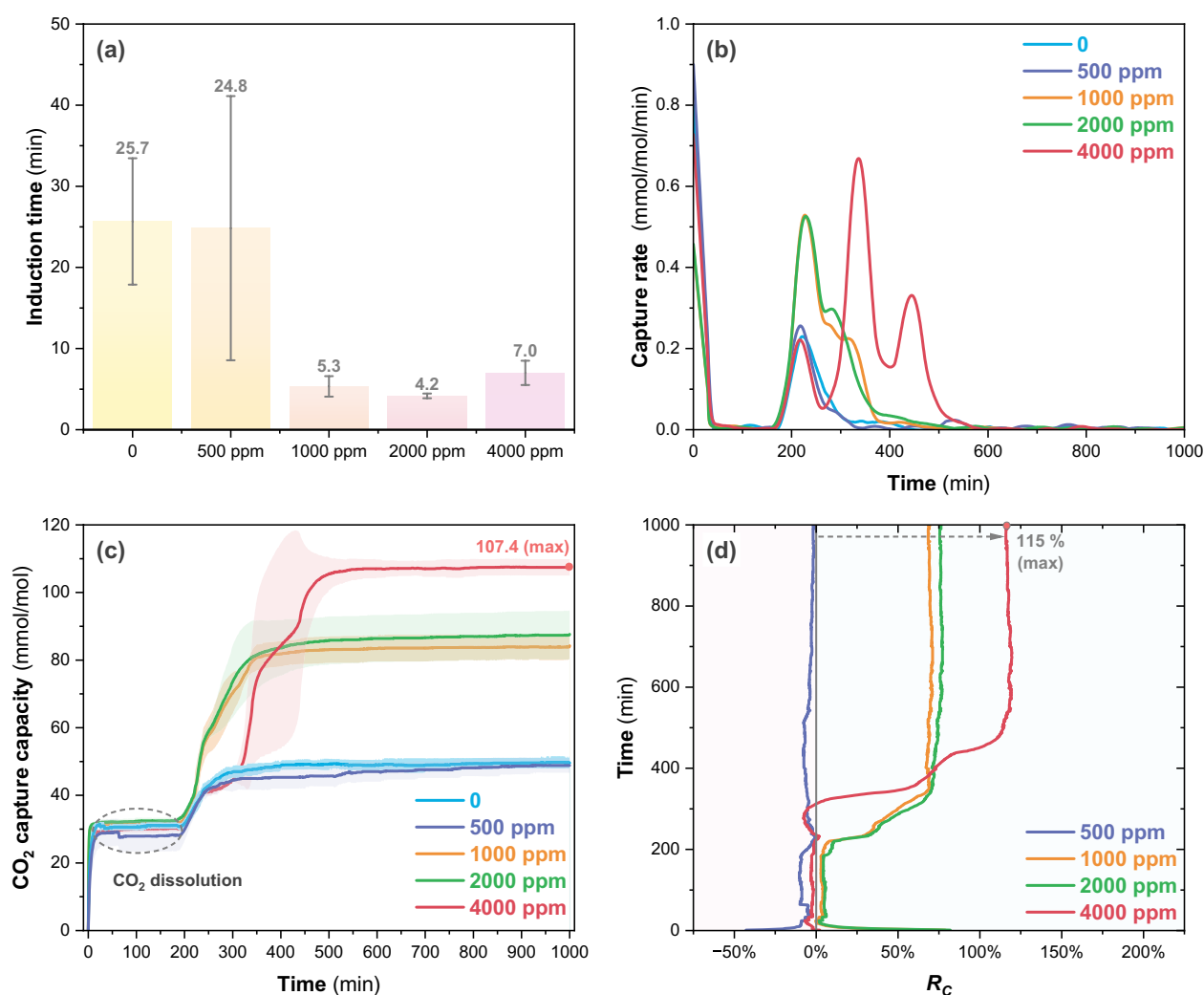


Fig. 6 CO₂ capture characteristics with strong mass transfer in the montmorillonite system via combining with SDS: (a) Induction time; (b) Capture rate; (c) CO₂ capture capacity; (d) Capture increase rate (Compared with Case 1–2)

local driving force and probability of hydrate nucleation, thereby shortening the induction time. However, further kinetic analysis based on Fig. 6b, c, and d reveals that the system with 500 ppm SDS shows little improvement. After a 180-min dissolution period followed by cooling, systems with SDS concentrations ≥ 1000 ppm consistently maintain higher CO₂ capture rates than both the low-concentration (500 ppm) SDS system and the pure mmt system. This phenomenon appears to be closely related to the critical micelle concentration (CMC) of SDS [50, 51]. In the mmt suspension, the 500 ppm SDS concentration likely remains below the CMC, limiting micelle formation and its promoting effect.

In addition, an interesting kinetic phenomenon was observed: when the SDS concentration increased from 1000/2000 ppm to 4000 ppm, the initial capture rate showed a decrease (Fig. 6c). Despite this slower onset,

the 4000 ppm system ultimately achieved the highest final CO₂ capture capacity of 107.4 mmol/mol, with experimental data stabilizing near the CO₂ phase equilibrium curve and approaching the theoretical maximum. This suggests that the higher SDS concentration provides a greater number of micelles and accommodation sites, leading to superior ultimate capture capacity. Nevertheless, excessive micelle formation may increase the initial viscosity of the suspension, which could adversely affect early-stage kinetics [52]. Furthermore, comparing these cases with Case 1–2 (1 wt% mmt) enables a clear observation of their dynamic enhancement (Fig. 6d). In summary, the addition of SDS at concentrations ≥ 1000 ppm significantly promotes both the CO₂ capture rate and the final capture capacity.

Furthermore, we tested pure mmt suspensions at different concentrations (0.5–5 wt%) under high stirring rate

(600 rpm) conditions and compared their capture performance with baseline cases under low mass transfer conditions (400 rpm). The experimental results demonstrate that under high mass transfer conditions, even without SDS compounding, the promoting effect of mmt itself on CO_2 capture is significantly amplified, exerting decisive enhancement on both key stages of hydrate formation—nucleation and growth. Firstly, during the nucleation stage, the high rotational speed creates intense disturbance at the gas–liquid interface, combined with the nucleation sites provided by mmt, resulting in shorter induction times at 600 rpm compared to equivalent concentrations at 400 rpm (Fig. 7a). Secondly, during the growth stage, the “film-breaking” effect of high rotational speed is sustained, achieving “stable hydrate film disruption”. This is clearly observable during the growth evolution process from 200–400 min (Fig. 7b). The specific “film-breaking” analysis is shown in the subsequent visualization section.

Regarding the final CO_2 capture capacity (Fig. 7c), all systems under high stirring rate reached relatively high levels, with the 0.5/1/3 wt% systems all outperforming pure water—this contrasts with the low rotational speed systems (Fig. 5c). Specifically, certain concentration mmt suspensions (e.g., 1 wt%) achieved capture capacities approaching or reaching 93.4 mmol/mol. Moreover, the intense turbulence prevents the complete formation and coverage of dense hydrate films on mmt particle surfaces, enabling gaseous CO_2 to continuously contact the liquid phase and active sites through dynamic gaps, thereby maintaining sustained rapid hydrate growth. Consequently, all 600 rpm cases demonstrated superior capture capacity compared to their corresponding 400 rpm cases (Fig. 7d), proving the fundamental role of high mass transfer conditions. It is worth noting that under high stirring rates, when mmt concentration is too low (0.5 wt%), the system lacks sufficient effective nucleation

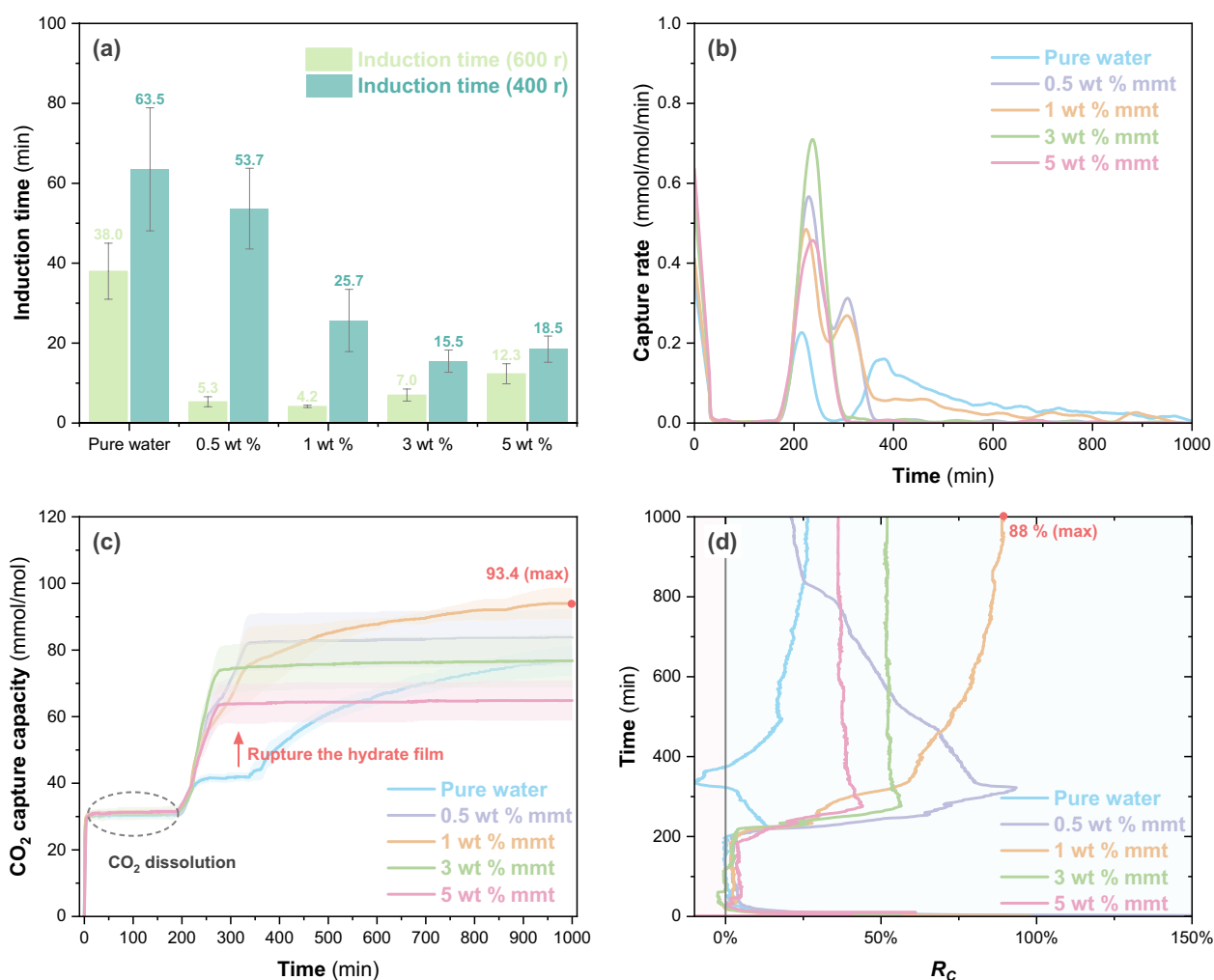


Fig. 7 CO_2 capture characteristics with strong mass transfer in the montmorillonite system via increasing stirring rate: (a) Induction time; (b) Capture rate; (c) CO_2 capture capacity; (d) Capture increase rate (Compared with the 400 r case of mmt at the same concentration)

sites; when concentration is too high (e.g., 5 wt%), the excessive suspension viscosity can diminish the mixing benefits brought by high mass transfer and may even encapsulate or mask some active sites, leading to reduced promotion effectiveness.

Figure 8 provides a visual comparison of the morphological evolution of CO₂ hydrate through the side and top view windows of the reactor under different mass transfer conditions and promoter formulations. These images offer compelling visual evidence supporting our preceding kinetic analysis. Here, $t=0$ represents 1 s before nucleation, $t=1$ s marks the nucleation instant, and other time points show characteristic moments after nucleation. Figure 8a reveals that at the nucleation moment with 1000 ppm SDS, the gas–liquid interface level drops abruptly, accompanied by massive hydrate formation, indicating nucleation sites primarily originating from bubbles induced by low-concentration SDS. This phenomenon is absent under high SDS concentration conditions (Fig. 8b), consistent with the formation rate patterns near this time point in Fig. 6b. However, such a massive instantaneous formation suggests possible rapid coverage of the interface by hydrate film. The top view (Fig. 8a) confirms the stable presence of hydrate film after nucleation, which impedes subsequent mass transfer, aligning with this system's non-optimal final capture capacity. Conversely, under high SDS concentration (Fig. 8b), the top view shows a discontinuous hydrate film with gaps, enabling continuous growth along the reactor wall.

Furthermore, this phenomenon (Fig. 8c and d) provides direct visual proof of the stable hydrate film disruption mechanism enabled by high stirring rate (600 rpm). During stable reaction stages, the gas–liquid interface isn't completely covered by a continuous hydrate film but rather shows clearly visible gaps in the CO₂ hydrate layer. The strong fluid shear forces generated by high stirring rates continuously disrupt the formation of a dense

hydrate layer, creating dynamic channels for sustained gas transport into the liquid phase. For lower concentration mmt suspensions (1 wt%, Fig. 8c), the system's lower viscosity allows more effective transmission of turbulent energy to the interface, resulting in larger and more stable gaps with reduced mass transfer resistance. This visual evidence correlates well with this case's fastest capture rate and highest capacity. When mmt concentration increases (5 wt%, Fig. 8d), the elevated suspension viscosity dampens turbulent energy, reducing the film-disruption efficiency and yielding smaller, less stable gaps. This intuitively explains why excessively high mmt concentrations, despite showing improved performance under high-speed stirring (Fig. 7d), ultimately underperform optimal lower-concentration cases—the constricted mass transfer channels (gaps) directly limit continuous CO₂ supply.

The aforementioned findings demonstrate that modifying the external experimental conditions—specifically through high mass transfer conditions (increased stirring rate) and the addition of SDS as two external kinetic enhancement strategies—significantly optimizes both the induction and growth processes of CO₂ hydrates in the mmt system, while effectively enhancing its final capture capacity.

3.3 Clay types and acid leaching modification

To further investigate the fundamental influence of intrinsic material properties on hydrate formation kinetics, this section shifts the research focus from external conditions to the material's internal structure. It aims to systematically explore two enhancement pathways based on microstructural modification of materials: first, by comparing illite, which possesses different interlayer structures and surface properties, to study the impact of clay mineral type; second, by performing acid leaching modification on mmt itself to regulate its specific surface

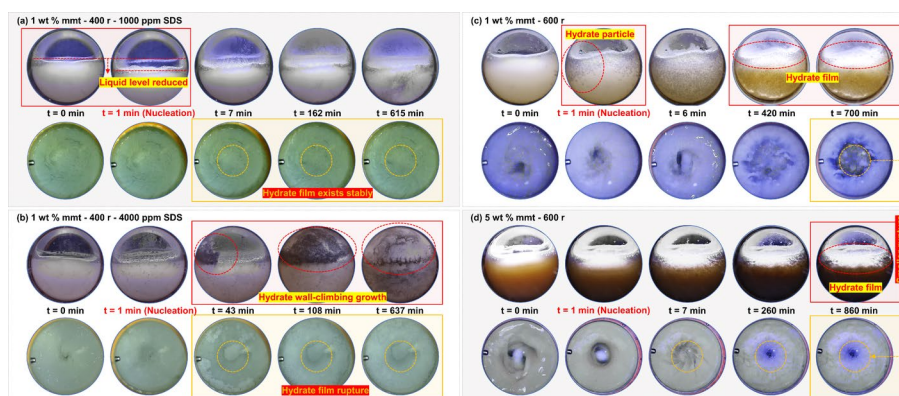


Fig. 8 CO₂ hydrate formation process with strong mass transfer in the montmorillonite system: (a) Case 2–2; (b) Case 2–4; (c) Case 3–2; (d) Case 3–4

area, and pore structure, thereby revealing the enhancement mechanism of material microstructure evolution on CO₂ hydrate formation kinetics.

Firstly, we systematically compared the performance of illite and mmt across different concentrations (0.5–5 wt%) under high mass transfer conditions (600 rpm). The experimental results demonstrate that illite, as a non-swelling clay (the expansion rate is only 1–2%), exhibits significantly weaker promotion effects overall compared to mmt with its expansive interlayer structure. As shown in the induction time data (Fig. 9a), all illite systems exhibited longer induction times than their mmt counterparts at equivalent concentrations. This directly confirms that mmt's unique expansive structure more effectively reduces the nucleation energy barrier, accelerating the formation of hydrate crystal nuclei. Observations from the capture rate curves (Fig. 9b) and final capture capacity (Fig. 9c) indicate that illite systems

only demonstrate modest promotion effects at low concentrations (e.g., 0.5–1 wt%), with both capture rates and ultimate capacities remaining lower than those of mmt systems at the same concentrations. The final R_C values (Fig. 9d) reveal that illite only slightly outperforms mmt at the high concentration of 5 wt%, while underperforming mmt systems at all other concentrations.

This discrepancy stems from fundamental differences in their microstructures (Fig. 10): (i) mmt possesses excellent layered expansibility and a larger specific surface area [53], with interlayer domains capable of accommodating water molecules and CO₂ gas, thereby creating abundant nanoconfined spaces that provide numerous efficient nucleation sites for hydrate formation; (ii) illite features tightly bound layers connected by potassium ions, classifying it as a non-expanding clay with substantially lower specific surface area and pore volume than mmt. Consequently, illite primarily

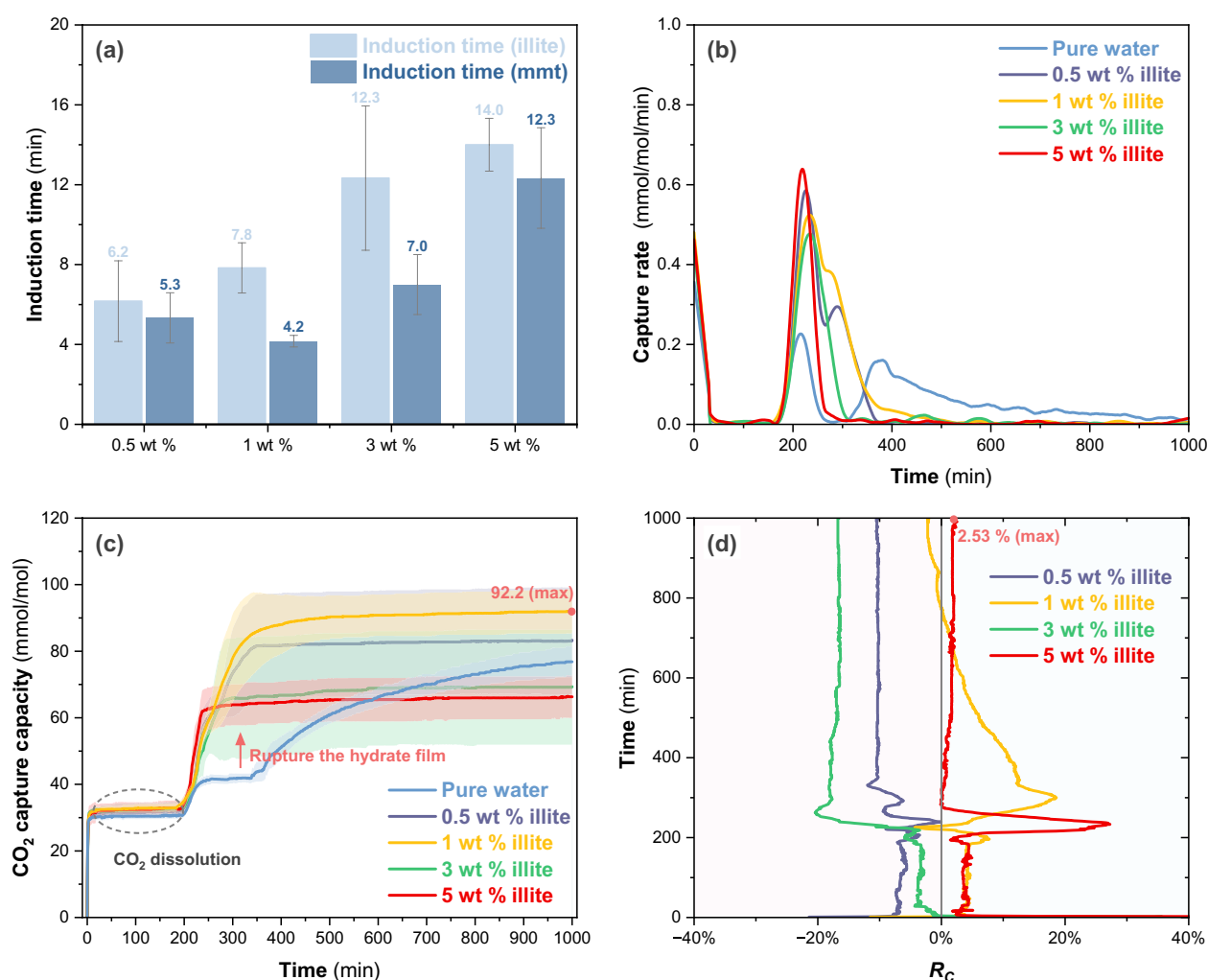


Fig. 9 CO₂ capture characteristics with various clays for illite system: (a) Induction time; (b) Capture rate; (c) CO₂ capture capacity; (d) Capture increase rate (Compared with the 600 r case of mmt at the same concentration)

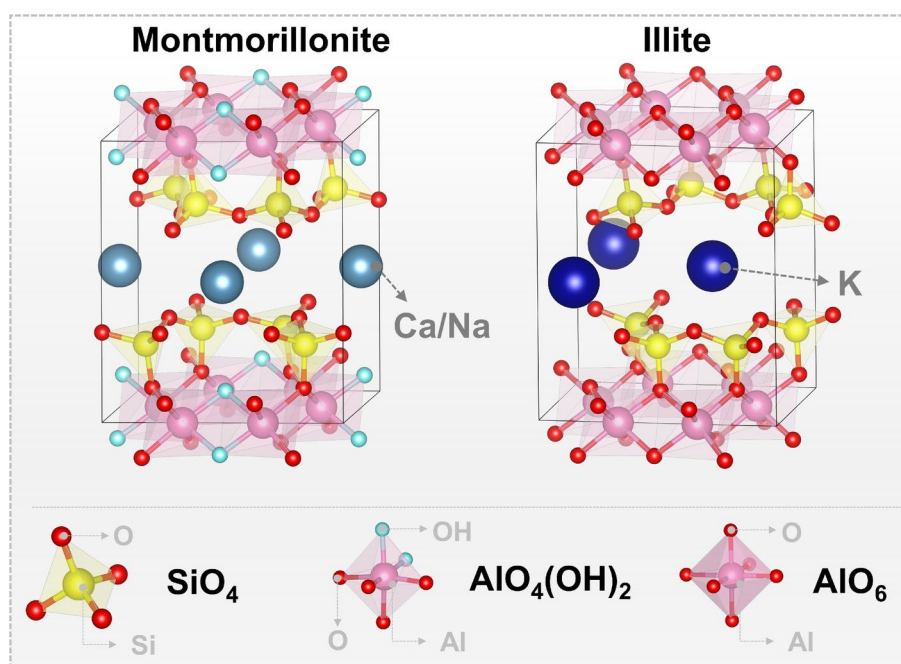


Fig. 10 Crystal structure of montmorillonite and illite

relies on external surfaces to provide nucleation sites, inherently limiting its promoting capability [19]. This comparative experiment clearly demonstrates that after high mass transfer conditions resolve external mass transfer limitations, the material's intrinsic physico-chemical properties (such as expansibility and specific surface area) become crucial determinants of its promotion capability. The inferior performance of illite conversely proves that expanding interlayer spacing, increasing specific surface area, and creating more nanopores are effective strategies for enhancing material performance. This provides compelling theoretical justification and experimental motivation for the subsequent acid leaching modification of mmt to further optimize its microstructure.

Subsequently, we conducted acid-leaching modification on mmt and systematically evaluated the promotion effect of the m-mmt under low mass transfer conditions (400 rpm). Characterization results (Fig. 3) confirmed that the acid modification successfully reconstructed the microstructure of mmt: SEM and PXRD revealed its transformation from a dense layered structure to a loose, porous amorphous structure. BET analysis (Table 3) further demonstrated that its specific surface area surged from $23.54 \text{ m}^2/\text{g}$ to $312.12 \text{ m}^2/\text{g}$, while the total pore volume significantly increased from $0.09 \text{ cm}^3/\text{g}$ to $0.73 \text{ cm}^3/\text{g}$, forming a more abundant hierarchical pore channel system.

The performance test results show excellent agreement with the characterization data, confirming the success of the modification strategy: (i) Significantly reduced induction time: As shown in Fig. 11a, the m-mmt systems demonstrated substantially shorter induction times across all tested concentrations compared to unmodified m-mmt. This can be directly attributed to the vastly increased specific surface area and abundant pore channels providing numerous active sites for CO_2 hydrate nucleation, thereby greatly reducing the nucleation energy barrier (Table 3). Hydrate nucleation tends to occur more readily in mesopores [54]. The hierarchical mesoporous structure of m-mmt provides abundant highly efficient nucleation sites for hydrate, while micropores mainly serve for the adsorption and enrichment of CO_2 , further increasing the local CO_2 concentration [55]. The synergistic effect of these two structures achieves dual enhancement of both nucleation and growth. (ii) Comprehensively enhanced capture kinetics: The capture rate curves in Fig. 11b and the dynamic increase comparison in Fig. 11d clearly show that the m-mmt systems maintained superior capture rates throughout the growth phase compared to the original mmt systems at corresponding concentrations. This improvement benefits from the developed porous network, where these channels serve as nucleation sites that effectively adsorb and concentrate CO_2 and water molecules, thus sustaining the high local reactant concentration required for rapid hydrate growth. (iii) Substantially improved ultimate capture capacity: According

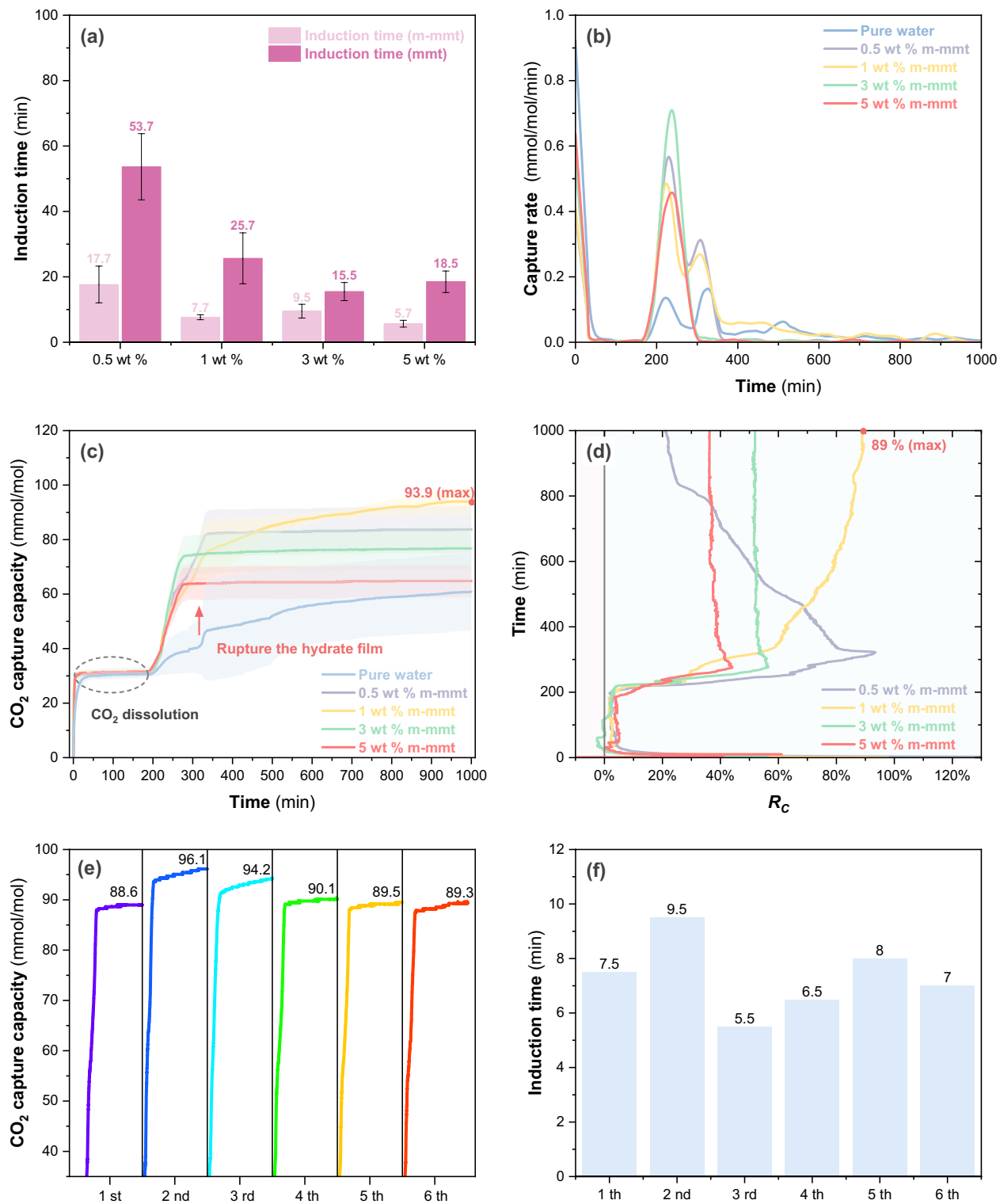


Fig. 11 CO₂ capture characteristics with various clays for the modified montmorillonite (m-mmt) system: (a) Induction time; (b) Capture rate; (c) CO₂ capture capacity; (d) Capture increase rate (Compared with the 400 r case of mmt at the same concentration); (e) CO₂ capture capacity in cyclic tests (1 wt% m-mmt); (f) Induction time in cyclic tests (1 wt% m-mmt)

to the final results in Fig. 11c and the R_C value changes in Fig. 11d, the m-mmt systems achieved comprehensive superiority in CO_2 capture capacity over the original mmt. Specifically, the 1 wt% m-mmt system reached a final capture capacity of 93.9 mmol/mol, representing an 83.9% increase compared to mmt under identical stirring rate and concentration conditions. (iv) Fig. 11e and f further reveal that at the optimal concentration of 1 wt%, the m-mmt system exhibits excellent recyclability, benefiting from its stable structure and other characteristics. It maintained a short induction time and high CO_2 capture capacity throughout 6 consecutive recyclability tests.

The underlying microstructural characteristics—the enormous specific surface area and well-developed hierarchical pore channels—not only determine the capture capacity but also profoundly influence the macroscopic growth morphology and kinetic processes of hydrates within the reactor, as illustrated in Figs. 12 and 13. Through continuous observation of hydrate growth patterns in different m-mmt systems, we discovered: (i) Fundamental transformation of hydrate growth mode: In the low concentration range (0.5–1 wt%), no significant amount of macroscopically visible suspended hydrate particles formed in the system. This does not indicate poor promotion effectiveness, but rather results from the ultra-high specific surface area of m-mmt that highly disperses its exceptional nucleation sites, promoting

hydrates primarily grow as “film-like” or “dispersed” forms attached to particle surfaces and within pore channels, rather than aggregating into large particles. (ii) Relatively complete hydrate reaction with notable wall-climbing phenomenon: Under m-mmt concentrations of 0.5/1/3 wt%, distinct hydrate wall-climbing appeared after complete reaction (Fig. 13, red boxes). As m-mmt concentration increased, CO_2 hydrate particles became increasingly dense in the side view (Fig. 12, red boxes), with the maximum particle concentration observed at 3 wt% m-mmt. From the top view, a cyclic phenomenon of hydrate film rupture-regeneration-rerupture was observed (Fig. 13d), which actually represents a visual manifestation of the non-compact hydrate growth pattern on m-mmt surfaces, creating sustainable mass transfer channels under low mass transfer conditions (400 rpm).

Under identical mass transfer conditions, the physico-chemical properties of the materials themselves become the dominant factor determining performance differences. Here, mmt, due to its unique expandable layered structure and larger specific surface area, performs significantly better than illite, which has tightly bound, non-expandable interlayers. This comparison strongly demonstrates that abundant nanoconfined spaces are crucial for efficient hydrate nucleation and growth. The acid leaching modification of montmorillonite successfully transformed its structure from dense and layered to loose, porous, and

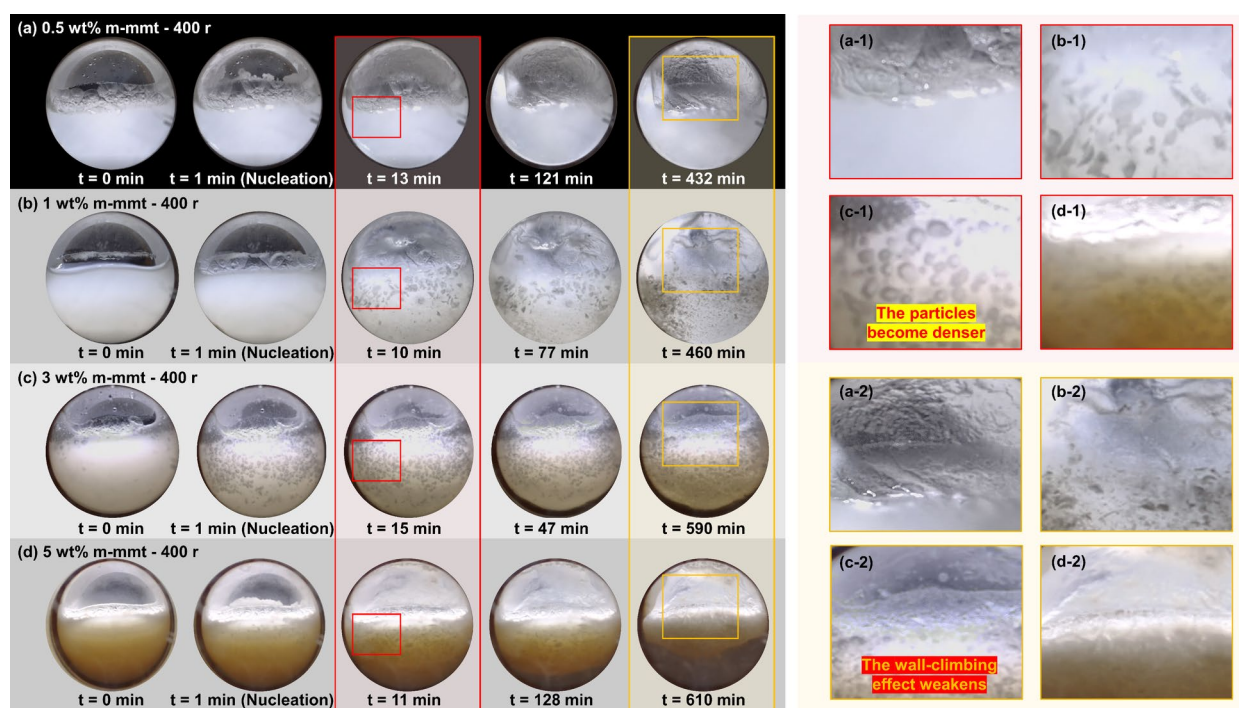


Fig. 12 CO_2 hydrate formation process for modified montmorillonite (m-mmt) system in the side window: (a) Case 5–1; (b) Case 5–2; (c) Case 5–3; (d) Case 5–4

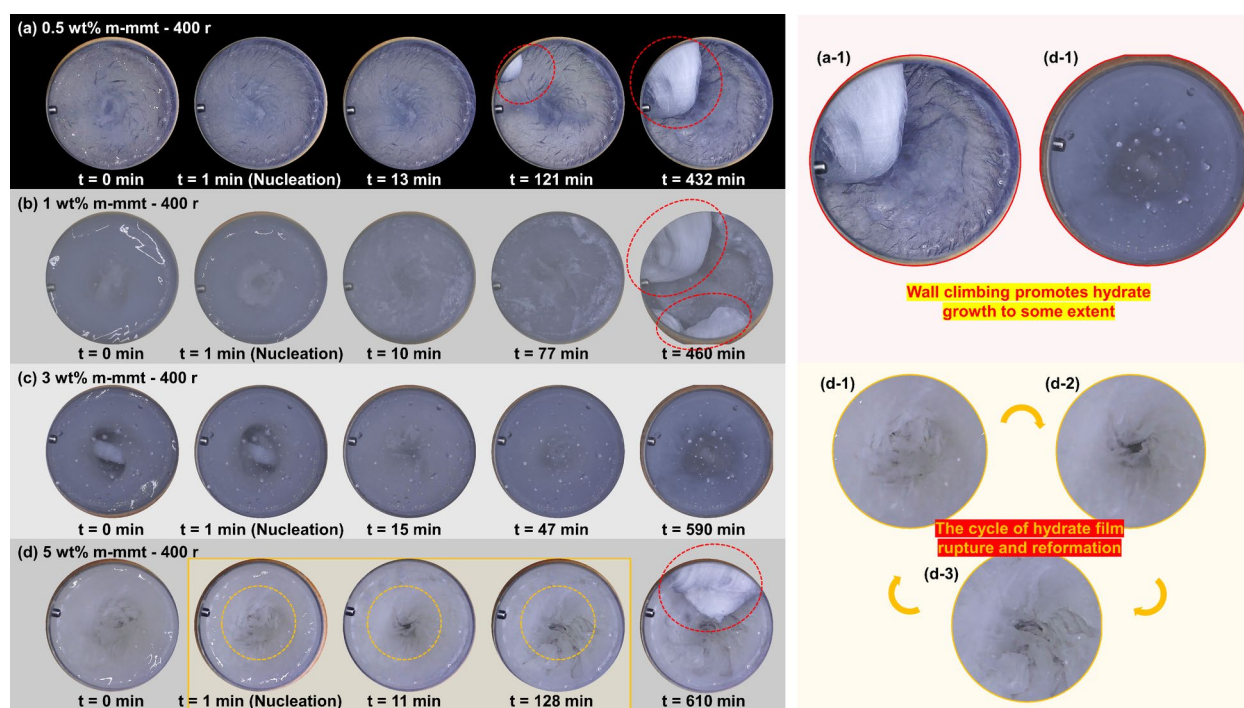


Fig. 13 CO₂ hydrate formation process for modified montmorillonite (m-mmt) system in the top window: (a) Case 5–1; (b) Case 5–2; (c) Case 5–3; (d) Case 5–4

amorphous. This restructuring enabled the m-mmt to simultaneously achieve the following under low mass transfer conditions (400 rpm): significantly shortened induction time, comprehensive acceleration of capture kinetics, substantial improvement in final capture capacity, and favorable regulation of growth morphology.

3.4 Discussion

3.4.1 CO₂ capture modes and capacity summary

Here, we further compiled the CO₂ capture capacities across all cases (Fig. 14). It is necessary to clarify

the apparent difference in dissolved CO₂ concentration between Figs. 5, 6, 7, 9, 11, and 14: the ~30 mmol/mol CO₂ dissolution shown in other figures is the peak value during the kinetic process, representing the maximum amount of physically dissolved CO₂ in the aqueous phase before massive CO₂ hydrate formation. In contrast, the ~13 mmol/mol dissolved CO₂ in Fig. 14 is the remaining stable value after all systems reach reaction equilibrium. The difference between them is approximately 17 mmol/mol, corresponding to the amount of dissolved CO₂ converted into CO₂ hydrate, accounting

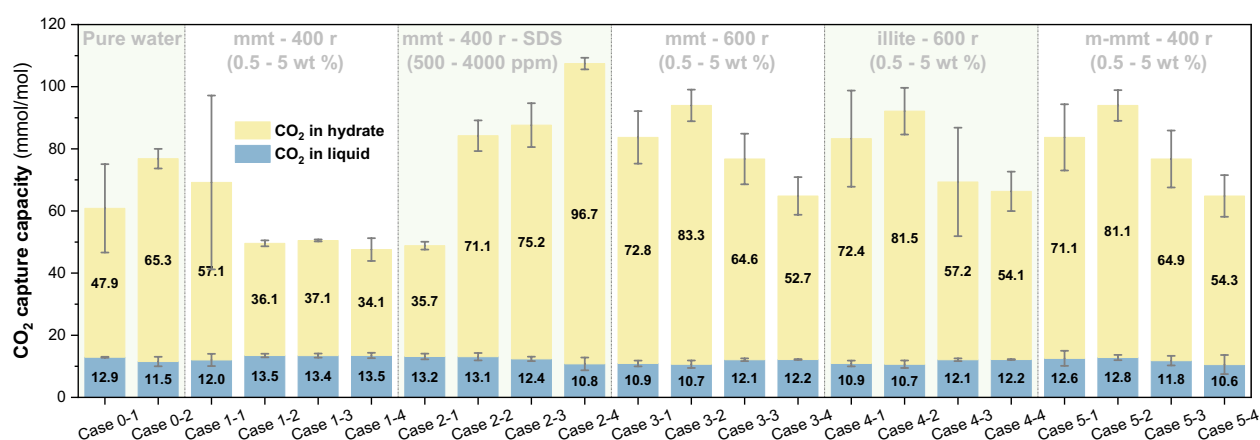


Fig. 14 Summary of CO₂ capture capacity in all cases (distinguishing between hydrate capture and dissolution capture)

for about 25% of the total CO₂ capacity bound in hydrates. This conversion further confirms that dissolved CO₂ in the aqueous phase is an important reactant for hydrate formation, and the nucleation sites provided by montmorillonite can effectively promote the conversion of dissolved CO₂ into hydrate crystals.

Further, it was observed that the amount of dissolved CO₂ remained relatively constant in all tested systems (pure water, mmt, illite, and m-mmt at different concentrations). This indicates that the primary mechanism of different promotion strategies is not significantly enhancing the physical solubility of CO₂, but rather improving the total capture capacity by altering the nucleation and growth kinetics of hydrates.

3.4.2 Formation mechanism of CO₂ hydrate

Figure 15 further clarifies the optimal pathways for performance enhancement, and Fig. 16 summarizes the schematic diagram of the synergistic enhancement mechanism for CO₂ hydrate formation. Overall, regarding the mmt usage strategy, low-concentration (0.5–1 wt%) clay systems generally demonstrated superior total CO₂ capture capacity. This holds true not only for mmt and m-mmt systems but also for illite, despite its weaker performance, with its best effects appearing within this concentration window. Low-concentration mmt provides abundant nucleation sites to shorten the induction time; the hydration of interlayer Ca²⁺ enhances gas–liquid contact and improves the hydrate growth rate (Fig. 16a). The hydration ability of K⁺, the charge-balancing ion in the illite interlayer, is weak, with only one layer of water molecules. This leads to effective interlocking between the illite layers (Fig. 16b), so that a card-house-like structure similar to that of montmorillonite cannot be formed.

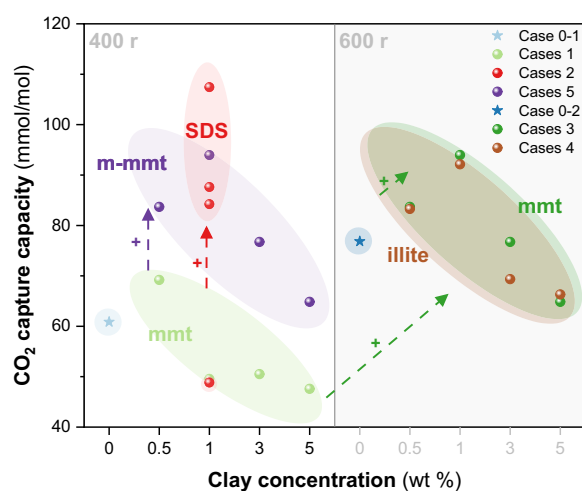


Fig. 15 Capture capability interval comparison

For external enhancement strategies, this study confirms their effectiveness and synergism. The compounding effect with SDS shows a positive correlation with its concentration, provided that it exceeds the critical micelle concentration to ensure the formation of solubilizing micelles (Fig. 16a1). Simultaneously, the generally superior performance of 600 rpm systems over 400 rpm systems visually demonstrates that improving mass transfer efficiency through intense disturbance to disrupt dense hydrate films is crucial for maintaining a continuous and rapid supply of CO₂ (Fig. 16a2).

Regarding the influence of the clay's inherent structure, the comparison with illite revealed the key roles of an expandable layered structure and high specific surface area. Building on this, the acid-leaching modification of mmt successfully maximized these two advantages. Acid leaching reconstructs the porous structure of mmt, increasing its BET surface area by 13-fold and forming hierarchical pores dominated by mesopores, which enhances CO₂ adsorption and enrichment, reduces the nucleation energy barrier, and enables efficient CO₂ capture with low energy consumption (Fig. 16c). As shown in Fig. 14, the acid-leaching modification enabled m-mmt under low mass transfer conditions (400 rpm) to achieve promotion performance or even reach that of unmodified mmt systems under high mass transfer conditions (600 rpm). For example, the capture capacity of 1 wt% m-mmt at 400 rpm surpassed that of mmt at the same concentration under both 400 rpm and 600 rpm conditions. This marks a successful transition in the promotion strategy: from forced enhancement relying on external energy input (high stirring rate) to intrinsic enhancement.

In addition, the nanoconfined interlayer space of clay significantly lowers the nucleation energy barrier of CO₂ hydrate by regulating molecular behavior and interfacial interactions, whose microscopic mechanism can be clarified by molecular simulation results. The surface hydroxyl and siloxane groups of mmt and illite can constrain water molecules and reduce the energy consumption for molecular rearrangement [56]. Interlayer cations such as Ca²⁺ and K⁺ form a hydration barrier to avoid excessive adsorption of CO₂ on the surface, while optimizing the tetrahedral arrangement of water molecules and further decreasing nucleation resistance [57]. After acid leaching modification, the crystal plane structure is disrupted, and the reduced hydroxyl groups weaken its hydrophilicity; the hydrophobic surface is more conducive to CO₂ adsorption and hydrate formation [58, 59]. In summary, the nanoconfined interlayer space of clay promotes the nucleation and growth of CO₂ hydrate both thermodynamically (lowering the energy barrier for molecular rearrangement) and kinetically (increasing

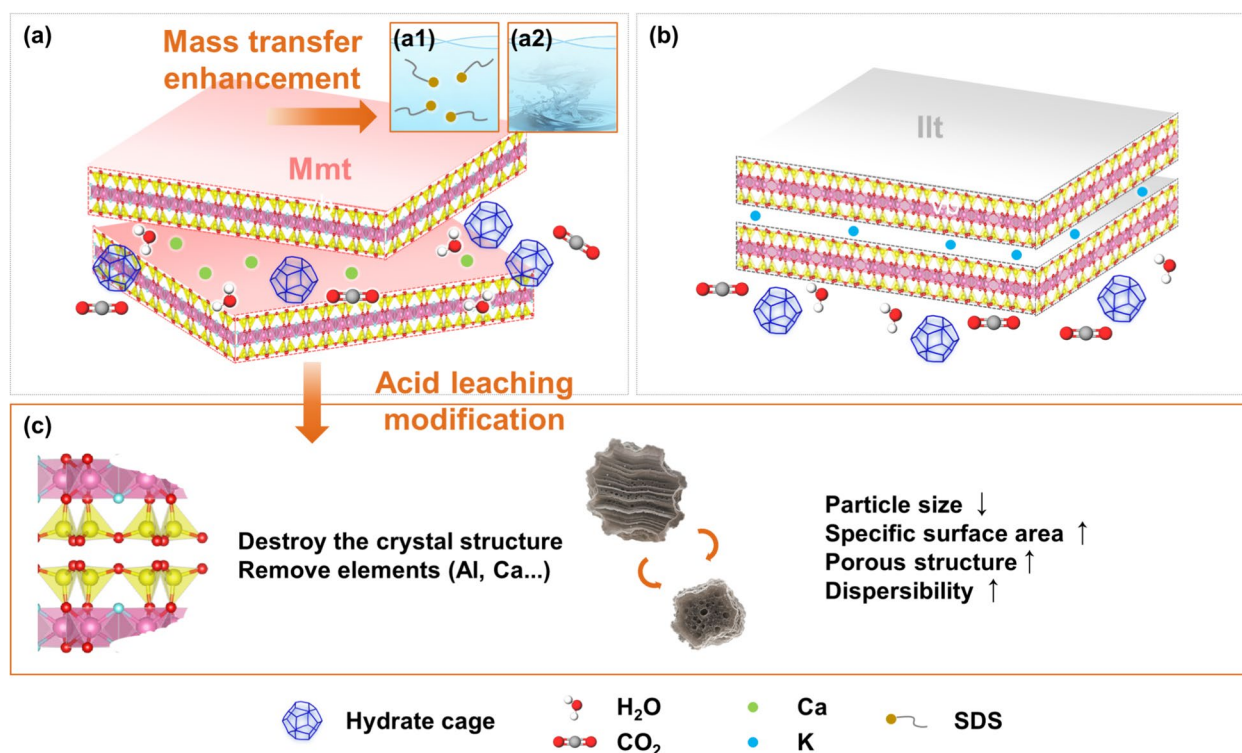


Fig. 16 The synergistic enhancement mechanism for CO₂ hydrate formation: (a) mmt system; (b) Illite system; (c) Acid leaching modification

local CO₂ concentration and accelerating mass transfer) via the multistage mechanism: surface charge regulation, molecular behavior confinement, and ion–interface synergy [60].

3.4.3 Economic cost-effectiveness analysis

Besides the above findings on kinetics and mechanism, common clay minerals are extremely low-cost for industrial applications [61], which makes the cost-effectiveness of mmt a core advantage supporting their practical application in hydrate-based CO₂ capture. On the one hand, unlike the widely studied novel synthetic nanomaterials for CO₂ capture (such as metal organic frameworks, graphene, etc.), which suffer from high preparation cost, difficulty in large-scale production, and complex synthesis processes [62], calcium-based mmt is naturally abundant clay minerals widely distributed on land, characterized by extremely low raw material cost and easy accessibility. Compared with other clay minerals (e.g., illite) and synthetic kinetic promoters, calcium-based mmt integrates multiple advantages: low raw material cost, simple modification process, low operating energy consumption, and excellent capture performance. Its acid-leached derivative m-mmt further amplifies these advantages through the reconstructed porous structure. This unique cost-performance ratio makes calcium-based montmorillonite a promising alternative to traditional synthetic promoters.

4 Conclusions

This study systematically investigates the enhancement of CO₂ hydrate formation kinetics using calcium-based montmorillonite (mmt) through external condition optimization and intrinsic material modification. The main findings are summarized as follows:

- (1) Clay minerals significantly enhance hydrate nucleation and growth, particularly at low concentrations. Compared to pure water systems, the addition of mmt provides abundant nucleation sites, substantially shortens the induction time, and increases the initial hydrate formation rate. In low-concentration suspensions (e.g., 0.5 wt%), clay particles improve agitation and mass transfer during the growth stage, achieving a CO₂ capture capacity of 69.2 mmol/mol, surpassing both pure water and high-concentration clay systems. These results demonstrate the feasibility of using clay minerals as effective kinetic promoters.
- (2) External mass transfer intensification and surfactant compounding are effective yet energy-intensive. Increasing the stirring rate to 600 rpm promotes continuous hydrate film disruption, achieving a maximum CO₂ capture capacity of 93.4 mmol/mol. Compounding mmt with SDS (≥ 1000 ppm) further enhances performance, reaching a notable capture capacity of 107.4 mmol/mol through micelle-mediated

ated local CO₂ enrichment. However, these strategies are limited by high energy input or increased viscosity at excessive SDS concentrations.

- (3) The expandable layered structure of mmt is crucial for its promoting effect. Comparative tests with non-expandable illite confirm that the high specific surface area and nanoconfined interlayer spaces of mmt are key to facilitating CO₂ hydrate nucleation and sustained growth. Acid-leaching modification achieves a leap in performance under mild conditions. The modified m-mmt exhibits a 13-fold increase in BET surface area and a porous amorphous structure. Using only 1 wt% m-mmt at 400 rpm, the system attains a CO₂ capture capacity of 93.9 mmol/mol, an 83.9% improvement over pristine mmt, outperforming even high-stirring systems and demonstrating a successful shift from “forced” to “intrinsic” enhancement.

In summary, while optimizing external conditions remains a valid approach, designing the microstructure of the promoter material itself emerges as a more fundamental and efficient strategy. The acid modification of mmt unlocks its full potential, enabling high-efficiency CO₂ capture with lower operational energy consumption. In addition, characterization results of modified m-mmt confirm its structural stability, with reduced interlayer cations and mild hydration reactions. These features prevent the degradation of its layered/porous structure and surface active sites, endowing it with considerable recyclability potential. Furthermore, compared to other materials such as MOFs, m-mmt exhibits distinct economic advantages. Its cost-effectiveness allows for prioritizing the full exertion of single-cycle capture efficiency without overemphasizing recyclability, which further enhances its practical application prospects in CCUS systems.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-026-00172-z>.

Supplementary Material 1.

Acknowledgements

This study has been partially funded by: (1) Shenzhen Science and Technology Program (Project No. JCYJ20240813112038050), (2) National Natural Science Foundation of China (Project No. 52404059), (3) the Economy, Trade and Information Commission of Shenzhen Municipality, China (Project No. HYCYPT20140507010002), (4) Shenzhen Key Laboratory of Advanced Technology for Marine Ecology, China (Project No. ZDSYS20230626091459009).

Authors' contributions

Hongyu Ye: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation. Peng Dong: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis. Mengya Niu: Visualization, Validation, Methodology, Data curation. Yuanxin Yao: Validation, Visualization. Daoyi Chen: Supervision, Resources,

Project administration, Funding acquisition. Jun Duan: Visualization. Mucong Zi: Writing – review & editing, Writing – original draft, Resources, Project administration, Conceptualization.

Data availability

Data will be made available on request.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Received: 17 December 2025 Revised: 5 March 2026 Accepted: 9 April 2026

Published online: 15 April 2026

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